

Rubric Summary – Calculus / Integration Exam

Q1 – 10 points (Integration & Substitution)

- Q1.1 – 3 points: Correct; integration error; minor computation error; incorrect.
- Q1.2 – 3 points: Correct; sign error; substitution error; incorrect; integration error after correct substitution.
- Q1.3 – 4 points: Correct; need to back-substitute; incorrect integral after correct substitution; incorrect.

Q2 – 10 points (Definite Integrals & Bounds)

- Q2.1 – 5 points: Correct; integration error; plug-in (evaluation) error; incorrect; sign error.
- Q2.2 – 5 points: Correct; sign error; one bound incorrect; incorrect; bounds incorrect; setup incorrect; incorrect.

Q3 – 10 points (Volumes: Cross-Sections, Shells/Washers)

- Q3.1 – 5 points: Correct; need to multiply by 2; missing square; incorrect; bounds incorrect; cross section incorrect.
- Q3.2 – 5 points: Correct; bounds incorrect; sign error; shell radius incorrect; incorrect application of washer/shell method.

Q4 – 10 points (Substitution, Formulas, Algebra)

- Q4.1 – 5 points: Correct; no bound change after substitution; minor computation error; incorrect; major computation error.
- Q4.2 – 5 points: Correct; no bound change after integration; setup/radius/integral incorrect; did not change bounds.

Q5 – 10 points (Applications/Constants & Distances)

- Q5.1 – 2 points: Correct; incorrect; unnecessary factor of pi; incorrect integral.
- Q5.2 – 4 points: Correct; computation error; incorrect method to compute k; incorrect.
- Q5.3 – 4 points: Correct; no distance factor; incorrect radius; incorrect bounds; incorrect.